SRI KRISHNA SARVEPALLI

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SUMMARY

MS in Electrical Engineering (San Jose State University, May 2026) focused on pre-silicon ASIC and SoC design verification and RTL design for CPU, cache, memory subsystem and interconnect IP. Built UVM environments with constrained-random stimulus, assertions and functional coverage for an AXI4 interconnect, a tri-speed Ethernet MAC and a multi-channel DMA engine, verified a 5-stage RV32I core in UVM against a C reference model, and designed an AXI4 MESI cache controller. Also built CDC and RDC analysis flows automated in Python and Tcl, and started a C++ and SystemC performance model of a systolic MAC array. Works in Synopsys VCS and Cadence Xcelium and debugs from waveforms and logs. Seeking an entry-level design verification or ASIC engineering role.

EDUCATION

Master of Science in Electrical Engineering, San Jose State University, San Jose, CA Aug 2024 - May 2026

Relevant Courses: Logic Verification with UVM, SoC Design and Verification with SystemVerilog, Advanced Computer Architecture

B.E. in Electronics and Communication Engineering, Visvesvaraya Technological University Aug 2019 - Jun 2023

TECHNICAL SKILLS

Verification and Debug: UVM (agents, drivers, monitors, sequencers, scoreboards), constrained-random and directed testing, reference-model checking, SystemVerilog Assertions (SVA), formal property verification, functional coverage and coverage closure, regression testing, waveform and log debugging, failure triage

RTL Design: Synthesizable SystemVerilog and Verilog, FSMs, datapaths, pipelining, FIFOs and async FIFOs, synchronizers, register interfaces

Protocols and Architecture: AXI4, AXI4-Lite, APB, I2C, Ethernet (IEEE 802.3), DMA with scatter-gather descriptors, 5-stage pipelines, set-associative caches, cache coherency (MESI), memory subsystems, systolic arrays, RISC-V (RV32I)

Static Sign-Off: RTL lint, clock domain crossing (CDC) analysis, reset domain crossing (RDC) analysis, violation triage, waiver development

Scripting and Automation: Python (regression and coverage automation), Tcl, Shell, Makefile, C, C++, SystemC, Git, Linux

EDA Tools: Synopsys VCS, Cadence Xcelium, Xilinx Vivado (ILA), Verilator

EXPERIENCE

Teaching Associate, CMPE 146 Real-Time Embedded System Co-Design, San Jose State University Jan 2026 - May 2026

- Taught embedded systems and RTOS in C and guided labs on microcontroller interfacing, debugging embedded C issues with students.

Design and Verification Trainee, Maven Silicon, Bengaluru, India May 2023 - Jun 2024

- Designed synthesizable RTL of an 8-bit packet router in Verilog that routes packets to one of three output channels by address.
- Built a SystemVerilog UVM testbench with constrained-random stimulus, assertions and functional coverage, closing coverage to 100%.

ACADEMIC PROJECTS

AMBA AXI4 Interconnect UVM Verification

- Built a UVM environment for an AXI4 interconnect that drives constrained-random traffic and checks ordering and arbitration behavior.
- Found 4 ordering and arbitration bugs, debugged each from waveforms and logs, and closed functional coverage to 100%.

Tri-Speed Ethernet MAC UVM Verification (10/100/1000 Mbps, IEEE 802.3)

- Built a UVM environment for a tri-speed Ethernet MAC across three clock domains (2.5, 25 and 125 MHz), covering the async FIFO crossings.
- Debugged 5 corner-case failures from waveforms and logs and reached 96% functional coverage.

Multi-Channel DMA Engine UVM Verification

- Built a UVM environment for a multi-channel DMA engine with scatter-gather descriptors over AXI4, with a reference model and assertions.
- Found and debugged races between concurrent channel transfers using waveform and log analysis.

RISC-V 5-Stage Pipelined CPU (RV32I): Design and UVM Verification

- Designed a 5-stage pipelined RV32I processor in SystemVerilog with hazard detection, operand forwarding, load-use stall and branch flush logic.
- Verified the core in UVM against a C instruction-set reference model with a randomized instruction-stream generator.
- Ported the core to an FPGA in Vivado with Python host scripts and debugged failures with ILA captures against simulation waveforms.

Set-Associative Cache Controller with MESI Coherency

- Designed a set-associative cache controller in SystemVerilog with an AXI4 memory-side interface, MESI state tracking and snoop handling.
- Verified hit, miss, eviction and snoop-driven MESI transitions with directed and constrained-random sequences and assertions.

CDC and RDC Sign-Off Flow for a Multi-Clock, Multi-Reset Subsystem

- Built a multi-clock subsystem with synchronizers, a Gray-coded async FIFO and a pulse synchronizer, seeded it with known CDC and reset violations, and wrote Python and Tcl scripts that parse reports and group violations by type.
- Triaged each violation as a real bug or safe crossing, fixed real ones in RTL, wrote justified waivers for the rest, and added reset assertions.

Systolic MAC Array Performance Model for AI Accelerators (C++, SystemC) (In Progress)

- Building a C++ and SystemC model of a systolic multiply-accumulate array to study throughput for AI accelerator workloads.
- Automating parameter sweeps in Python to cross-check simulated results against an analytical throughput model.